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MZ 505
Inventory Systems

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Session 12: The Order-up-to Inventory Model

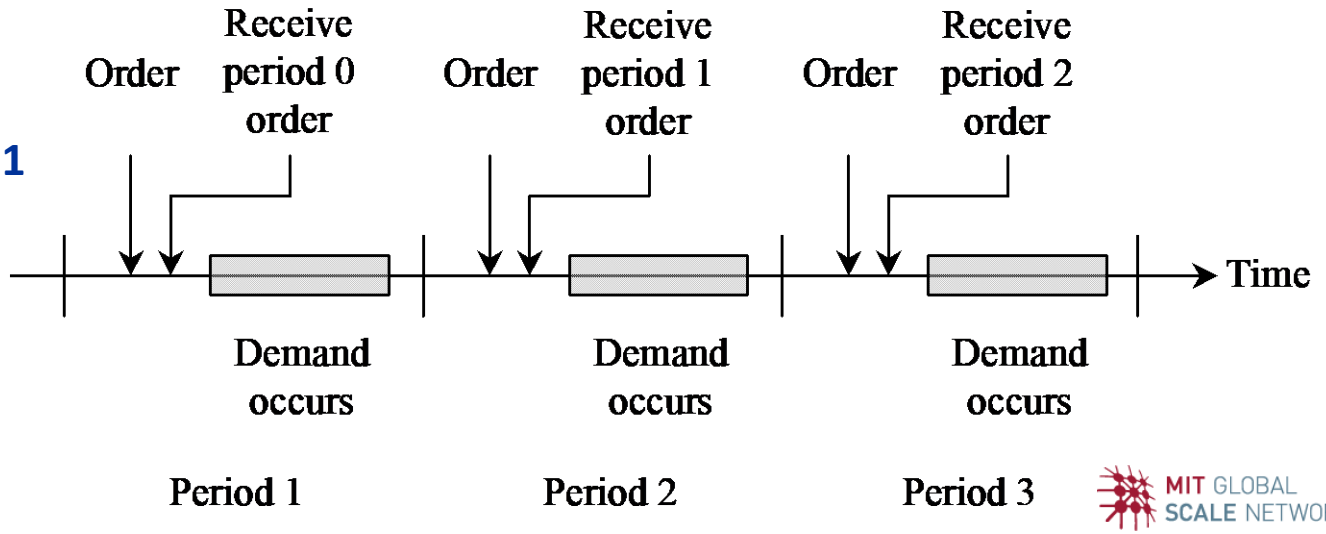
Adapted from Cachon and Terwisch slides (2013)

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Timing in the order-up-to model

- Time is divided into periods of equal length, e.g., 1 day, 1 week, etc.
- During a period the following **sequence of events** occurs:
  - A replenishment order can be submitted
  - Inventory is received
  - Random demand occurs
- Lead times (L):
  - An order is received after a fixed number of periods (=lead time)

An example with L = 1



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Order-up-to model definitions

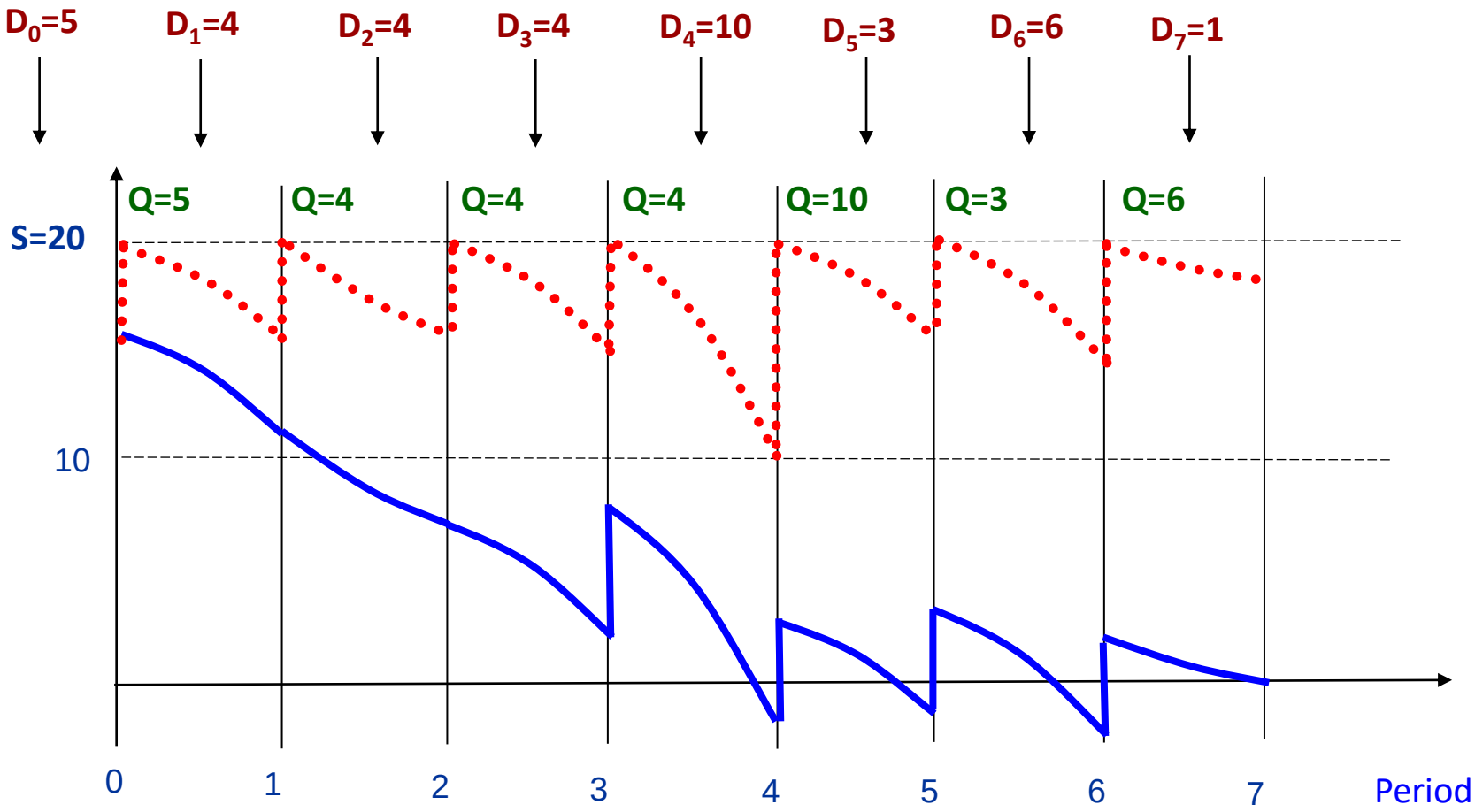
- **On-order inventory / pipeline inventory** = the number of units that have been ordered but have not been received
- **On-hand inventory** = the number of units physically in inventory ready to serve demand
- **Backorder** = the total amount of demand that has not been satisfied:
  - All backordered demand is eventually filled, i.e., there are no lost sales
- **Inventory level** = On-hand inventory - Backorder
- **Inventory position** = On-order inventory + Inventory level
- **Order up-to level, S**
  - The maximum inventory position we allow
  - Sometimes called the **base-stock level**

Order-up-to model implementation

- Each period's order quantity = S – Inventory position
  - Suppose S = 4
    - If a period begins with an inventory position = -3, how many units will be ordered? **7 units**
    - If demand were 10 in period 1, then how many units are ordered at the start of period 2? **10 units**

- A period's order quantity = the previous period's demand:
  - The order-up-to model is a pull system because inventory is ordered in response to demand
  - The order-up-to model is sometimes referred to as a 1-for-1 ordering policy

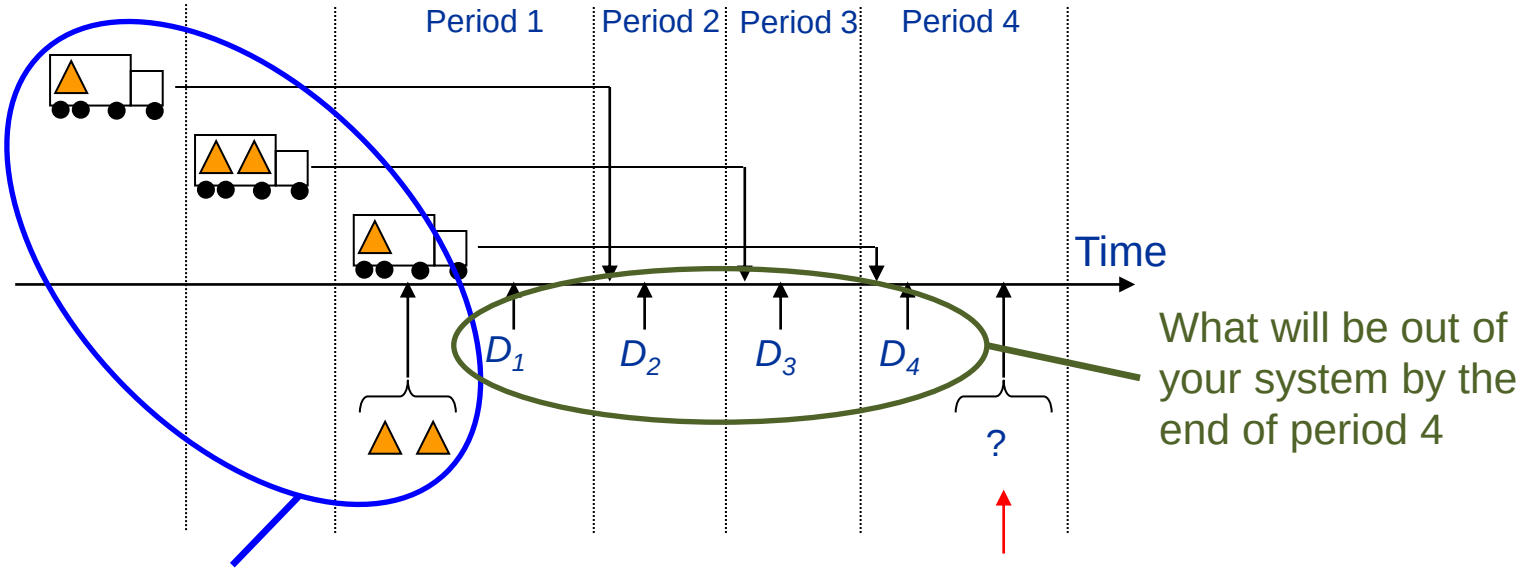
Example: L = 3, S = 20 and starting inventory (t_0) = 15



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Inventory level at the end of a period?

- Short answer:
  - Inventory level at the end of a period = S – (Demand over L+1 periods)
- Explanation via an example with *S* = 6, L = 3, and 2 units on-hand at the start of period 1



What will be (have been) in your system "physically" from start of period 1 until start of period 4 (=S)

Inventory level at the end of period 4 (this determines your costs)
= 6 - D_1 – D_2 – D_3 – D_4

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The Medtronic's InSync Pacemaker Case

Medtronic's InSync pacemaker SC



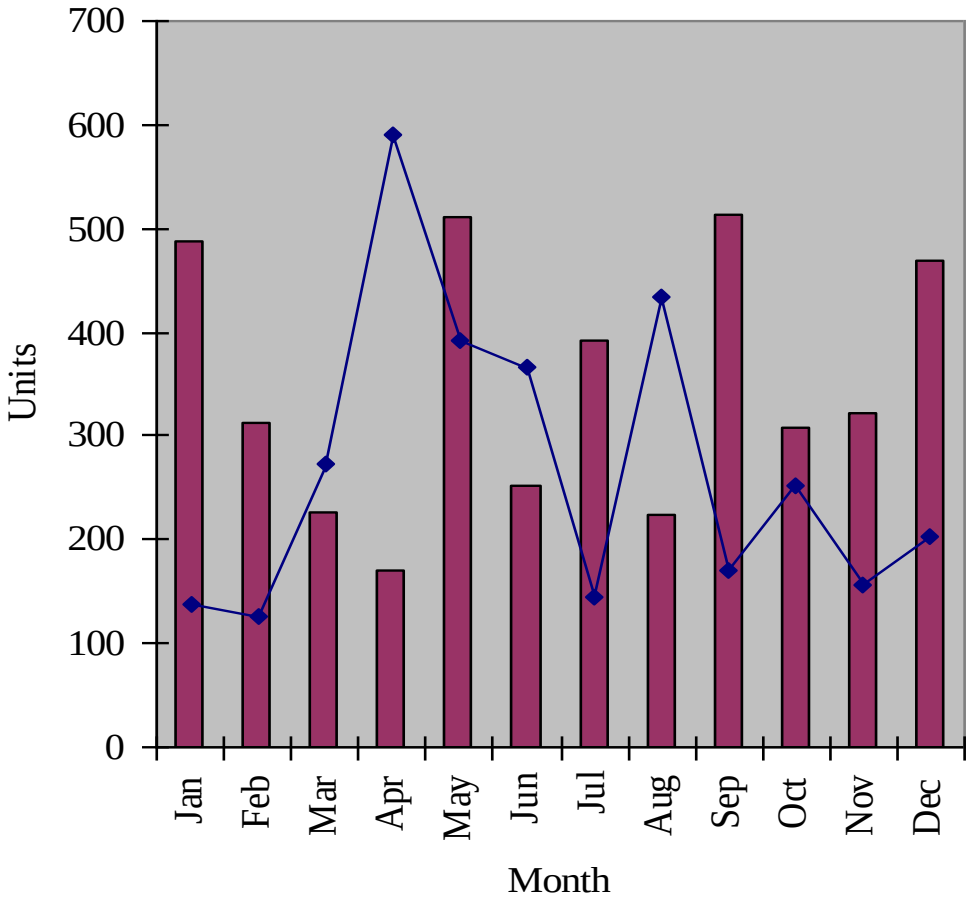
- InSync pacemaker supply chain:
  - One distribution center (DC) in Mounds View, MN
  - About 500 sales territories throughout the country
  - Consider Susan Magnotto's territory in Madison, Wisconsin

Medtronic: Management of field inventory

- Patients and surgeons do not tolerate backorders
- The pacemaker is small and has a long shelf-life
- Sales incentive system
- Each representative is given a par level which is set quarterly based on previous sales and anticipated demand

Objective: Because the gross margins are high, Medtronic wants an inventory control policy to minimize inventory investment while maintaining a very high customer service level.

InSync demand and inventory at the DC

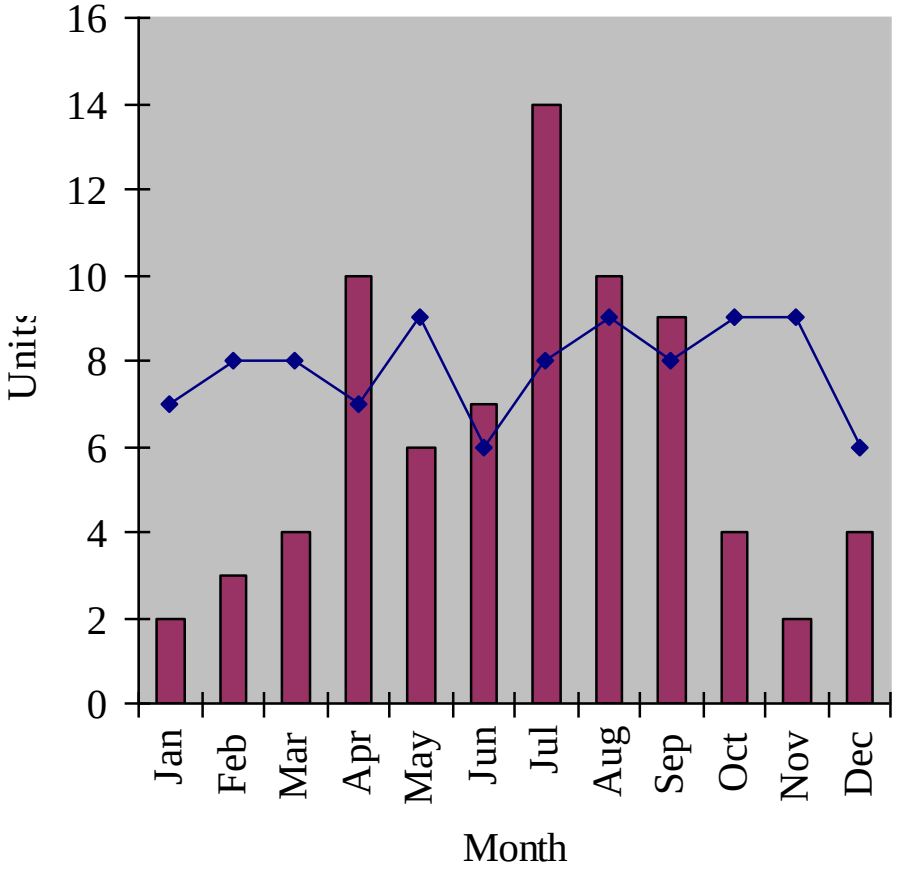


Monthly implants (bars) and end of month inventory (line)

Average monthly demand = 349 units

Standard deviation of monthly demand = 122.4

InSync demand and inventory in Susan's territory



Monthly implants (bars) and end of month inventory (line)

Total annual demand = 75 units

Average daily demand = 0.29 units (75/260), assuming 5 days per week

The Poisson demand distribution works better for slow moving items

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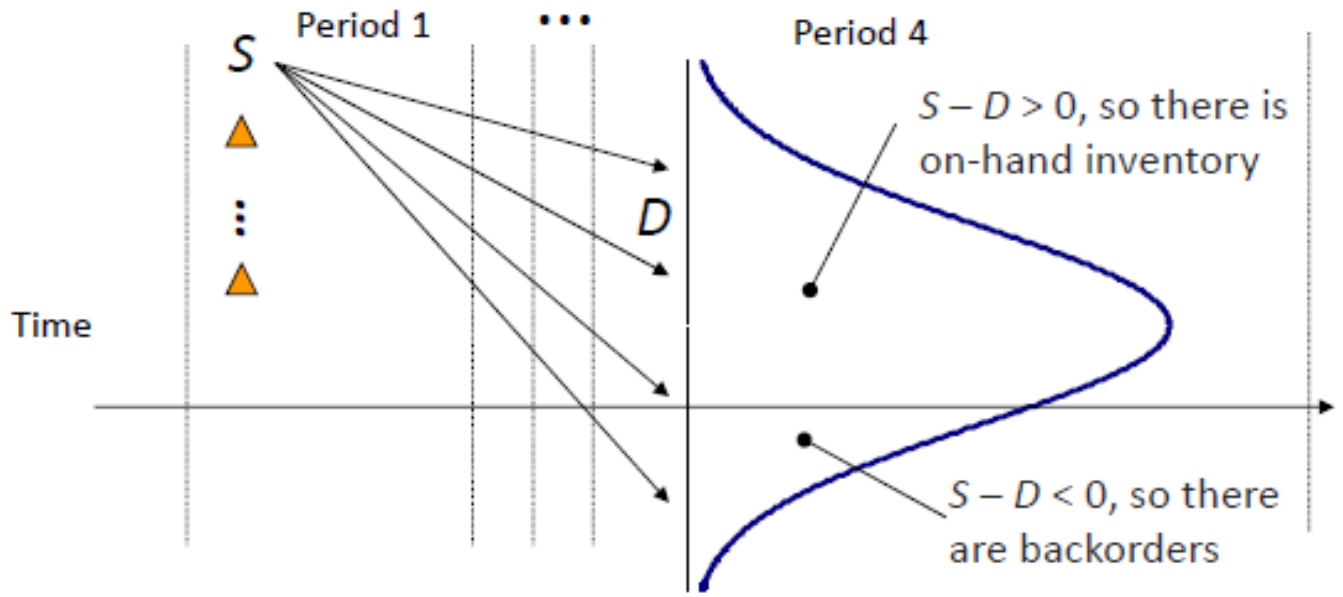
The Order-up-to Model: Performance measures

Stockout and in-stock probabilities

- The **stockout probability** is the probability at least one unit is backordered in a period
- The **in-stock probability** is the probability all demand is filled <u>in a period</u>:

In-stock probability = 1 – Stockout probability

= Prob{Demand over (L+1) periods ≤ S}

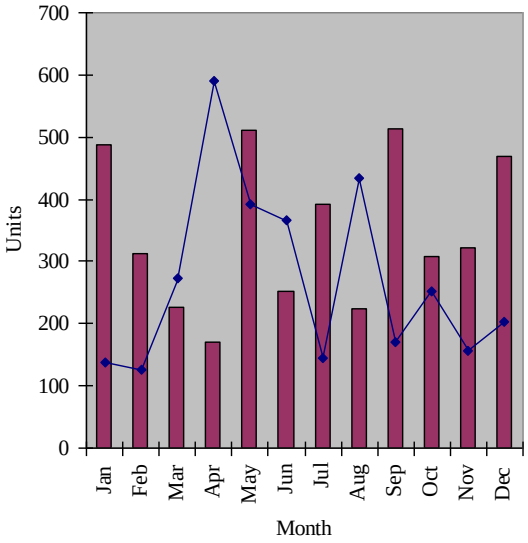


D = demand over L+1 periods

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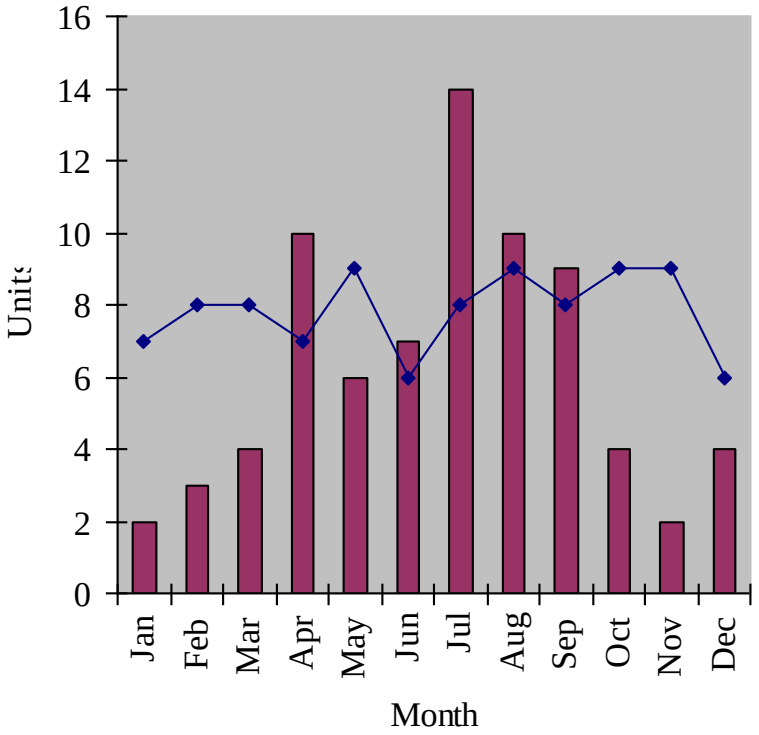
Demand over L+1 periods at the DC

- The period length is 1 week and the DC's lead time is 3 weeks
- Monthly demand is normally distributed with…
  - Mean = 349
  - Standard deviation = 122.4
- Weekly demand is normally distributed with…
  - Mean = 349 / 4.33 = 80.6
  - Standard deviation = 122.4 / sqrt(4.33) = 58.8
- L+1 demand is normally distributed with…
  - Mean = 80.6 x (3 + 1) = 322.4
  - Standard deviation = 58.8 x sqrt(3 + 1) = 117.6



Demand over L+1 periods in Susan's Territory

- The period length is 1 day, and Susan's replenishment lead time is 1 day
- Daily demand is Poisson with…
  - Mean = 0.29
- L+1 demand is Poisson with…
  - Mean = (1+1) x 0.29 = 0.58



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DC's Expected backorder assuming S = 625

- **Expected backorder** is analogous to the "Expected lost sales" in the Newsvendor model:
  - Normalize the order-up-to level S = 625:

$$z = \frac{S - \mu_{L+1}}{\sigma_{L+1}} = \frac{625 - 322.4}{117.6} = 2.57$$

  - Lookup L(z) in the Standard Normal Loss Function Table: L(2.57)=0.0016
  - Convert expected lost sales, L(z), for the standard normal into the expected backorder with the actual normal distribution that represents demand over L+1 periods:

$$E[\text{backorder}] = \sigma_{L+1} L(z) = 117.6 \cdot 0.0016 = 0.19$$

  - Therefore, if S = 625, then on average there are 0.19 backorders at the end of any period at the DC.

Other DC performance measures with S = 625

- **Expected on-hand inventory (or "Safety Stock")** is analogous to "Expected left-over" inventory in the Newsvendor model:

$$\text{E[on-hand]} = S - \text{E}[D_{L+1}] + \text{E[backorder]}$$
$$= 625 - 322.4 + 0.19 = 302.8$$

  - So, on average, there are 302.8 units on-hand at the end of a period

- **Expected on-order (pipeline) inventory** is the average number of units on order at any given time:

$$\text{E[on-order]} = \text{Lead Time} \times \text{E[Demand in one period]}$$
$$= 3 \cdot 80.6 = 241.8$$

- **Question:** How much is the **Expected cycle inventory? 80.6/2**

Other DC performance measures with S = 625 (cont.)

- **Fill rate (or expected fill rate)** is the fraction of customer demand per period satisfied from product in inventory:

$$\text{Fill Rate} = \frac{\text{E[demand satisfied from inventory]}}{\text{E[demand in one period]}} = 1 - \frac{\text{E[backorder]}}{\mu}$$

$$= 1 - \frac{0.19}{80.6} = 99{,}76\%$$

  - Stockout occurs when the demand during L+1 periods exceeds the order-up-to level, S
  - Consequently, while D_{L+1} is the basis for the calculation of E[backorder], the denominator in the fill rate considers the demand per period.

Performance measures in Susan's territory

	Mean demand = 0.29			Mean demand = 0.58	
S	$F(S)$	$L(S)$	S	$F(S)$	$L(S)$
0	0.74826	0.29000	0	0.55990	0.58000
1	0.96526	0.03826	1	0.88464	0.13990
2	0.99672	0.00352	2	0.97881	0.02454
3	0.99977	0.00025	3	0.99702	0.00335
4	0.99999	0.00001	4	0.99966	0.00037
5	1.00000	0.00000	5	0.99997	0.00004

$F(S) = Prob\ \{Demand\ is\ less\ than\ or\ equal\ to\ S\}$

$L(S)$ = loss function = *expected backorder* = expected amount demand exceeds S

- Suppose Susan operates with $S = 3$:
  - *What is the expected backorder?* 0.00335
  - *What is the in-stock probability?* 99.7%
  - *What is the expected on-order inventory?* 0.29
  - *What is the expected cycle inventory?* 0.29/2
  - *What is the expected on-hand inventory?* 3 – 0.58 + 0.00335
  - *What is the expected fill rate?* 1 – 0.00335/0.29

The Order-up-to Model:

Choosing an order-up-to level, S, to meet an in-stock service target

Choosing S to meet a target in-stock with normally distributed demand

- Suppose the target in-stock probability at the DC is 99.9%; that is, Prob{Demand over (L+1) periods ≤ S} = 0.999 :
  - From the Standard Normal Distribution Function Table, F(3.08)=0.9990
  - So we choose z = 3.08
  - To convert z into an order up-to level:

$$S = \mu_{L+1} + z \cdot \sigma_{L+1}$$
$$= 322.4 + 3.08 \cdot 117.6 = 685$$

  - Note that μ and σ are the parameters of the normal distribution that describes demand over L+1 periods

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Choosing S to meet a target in-stock with Poisson demand

	Mean demand = 0.29		Mean demand = 0.58		
S	$F(S)$	$L(S)$	S	$F(S)$	$L(S)$
0	0.74826	0.29000	0	0.55990	0.58000
1	0.96526	0.03826	1	0.88464	0.13990
2	0.99672	0.00352	2	0.97881	0.02454
3	0.99977	0.00025	3	0.99702	0.00335
4	0.99999	0.00001	4	0.99966	0.00037
5	1.00000	0.00000	5	0.99997	0.00004

$F(S)$ = *Prob* {*Demand is less than or equal to S*}

$L(S)$ = loss function = e*xpected backorder* = expected amount demand exceeds S

- *What S should Susan choose to yield an in-stock probability of at least 99%?* 3
- *What if the target is 99.9%?* 4
- *What if the target is 99.99%?* 5

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Justifying a service level via cost minimization

- Let h equal the holding cost per unit per period
- Let b equal the penalty per unit backordered
- "Too much-too little" challenge:
  - If S is too high, then there are holding costs, $C_o = h$
  - If S is too low, then there are backorders, $C_u = b$
- Cost minimizing order up-to level satisfies

$$\text{Prob}\{D_{L+1} \leq S\} = \frac{C_u}{C_u + C_o} = \frac{b}{b+h} \quad \text{(critical ratio)}$$

  - Suppose the review period is one week, the holding cost per unit per week is \$0.5 and the cost (penalty) per unit backordered is \$20
  - Optimal in-stock probability is 20/(20+0.5)=97.56%

Question

- Suppose in the order-up-to model the target in-stock probability is 0.95 and demands across periods are independent and identically Normally distributed.
- If the lead time is doubled but the target in-stock probability remains 0.95, what happens to the order-up-to level?

a. It more than doubles (increases by more than a factor of 2)

b. It doubles (increases exactly by a factor of 2)

c. It increases, but by less than a factor of 2

d. It decreases

e. The answer cannot be determined with the given information

The Order-up-to Model: General case

Periodic Review Policy: OUT-level and safety Inventory

- Consider the general case where:
  - T : Reorder interval
  - L : Lead time
  - μ, σ : mean and standard deviation of demand per unit time (assumed normally distributed)

- Then, the order-up-to level and safety cost calculations can be generalized as follows:

$$\mu_{L+T} = (L + T) \cdot \mu$$

$$\sigma_{L+T} = \sqrt{L + T} \cdot \sigma$$

$$S = \mu_{L+T} + z \cdot \sigma_{L+T}$$

$$E[\text{backorder}] = \sigma_{L+T} L(z)$$

$$\text{Safety Stock} = S - \mu_{L+T} + E[\text{backorder}]$$

Example: Periodic review policy

- μ = 2,500/week; σ = 500 (normally distributed)
- L = 2 weeks; T = 4 weeks; CSL = 0.90
- What is the required order-up-to level and safety inventory?
  - We have:

$$\mu_{L+T} = (L + T) \cdot \mu = 15{,}000$$

$$\sigma_{L+T} = \sqrt{L + T} \cdot \sigma = 1224.7$$

$$S = \mu_{L+T} + z \cdot \sigma_{L+T} = 15000 + 1.2815 \cdot 1224.7 = 16{,}570$$

$$E[\text{backorder}] = \sigma_{L+T} L(z) = 1224.7 \cdot 0.0473 = 53.25$$

$$\text{Safety Stock} = S - \mu_{L+T} + E[\text{backorder}] = 1{,}623.25$$

The Order-up-to Model:

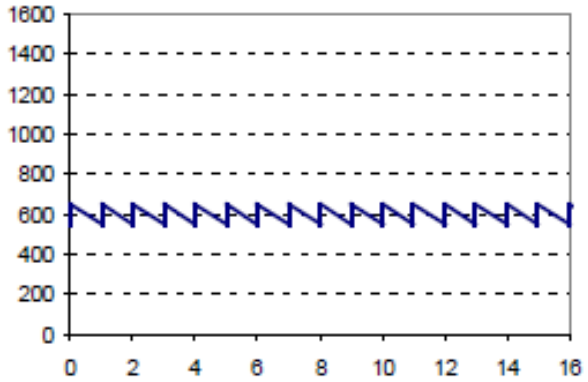
Balancing ordering & holding costs

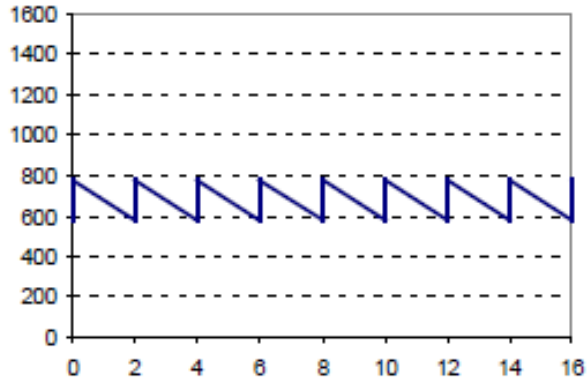
Impact of review period length

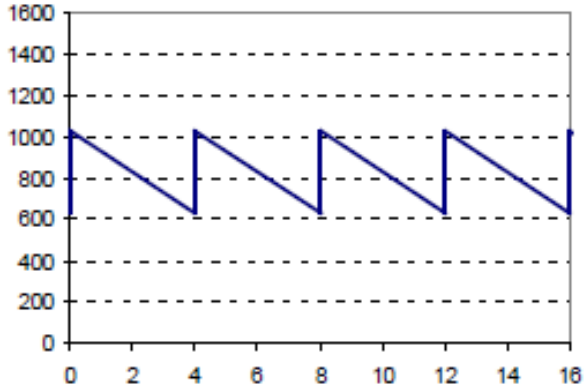
- Increasing the period length leads to larger and less frequent orders:
  - The average order quantity = expected demand in a single period
  - The annual number of orders *approximately* equals 1 / (length of period in years)
- Suppose there is a cost to hold inventory and a cost to place an order (independent of the quantity ordered)…
- … then there is a tradeoff between carrying little inventory (favored by short period lengths) and reducing ordering costs (favored by long period lengths)

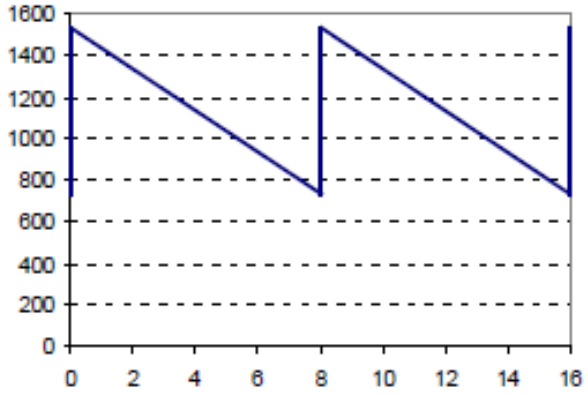
Example with mean demand per week = 100 and standard deviation of weekly demand = 75

- Inventory over time follows a “saw-tooth” pattern
- Period lengths of 1, 2, 4 and 8 weeks result in average inventory of 597, 677, 832 and 1130 respectively:









Assumption:
Inventory ownership passes over upon order delivery

Cycle inventory

Safety stock

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Impact of review period on total costs

- Assume there is a fixed cost associated with placing an order (i.e., this cost is independent of the quantity ordered)
- Let n be the number of orders in a year; this is given by:

$$n = \frac{1 - \text{Prob}\{\text{Demand in one period} = 0\}}{\text{Length of review period (in years)}}$$

- The average inventory per unit of time is given by:

$$\text{Average Inventory} = \text{safety stock} + \text{average cycle inventory}$$

- The "optimal" review period is the one that minimizes the sum of ordering and holding cost per unit of time (e.g., per year)
  - More on this in the solution of your Quiz Problem

Quiz problem (d): Lead time = 4 days

	Period Length (in days)			
	1	2	3	4
Expected demand over L+T				
Standard deviation over L+T				
Target in-stock probability				
z-statistic				
Order-up-to level, S				
Expected backorder				
Expected on-hand inventory (safety stock)				
Expected demand over review period				
Average cycle inventory				
Average inventory (per day)				
Annual holding cost				
Number of orders per year				
Ordering cost per year				
Annual ordering and holding cost				

The Order-up-to Model: Managerial insights

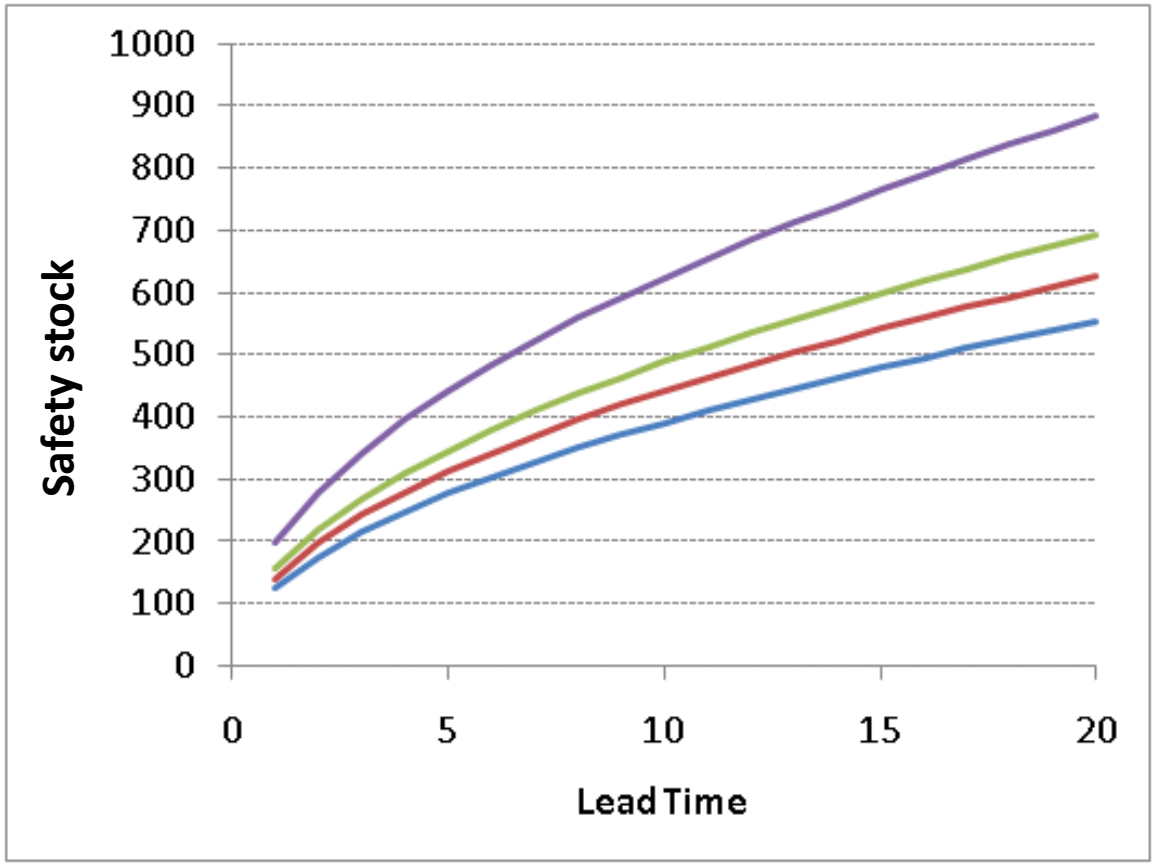
What determines the average safety stock?

- Using the performance metric equations discussed, we can derive an equation for the days of supply of on-hand inventory (or safety stock) for normally distributed demand, which has three components:

$$\text{Days of supply} = \underbrace{\sqrt{L+1}}_{\text{Lead time}} \times \underbrace{\left(\frac{\sigma}{\mu}\right)}_{\text{Variability of demand}} \times \underbrace{(\varphi(z) + z\Phi(z))}_{\text{In-stock/service level}}$$

  - L = lead time
  - σ = standard deviation of one period demand
  - μ = mean of one period demand
  - z = z-statistic
  - φ(z), Φ(z) = standard normal pdf and cdf

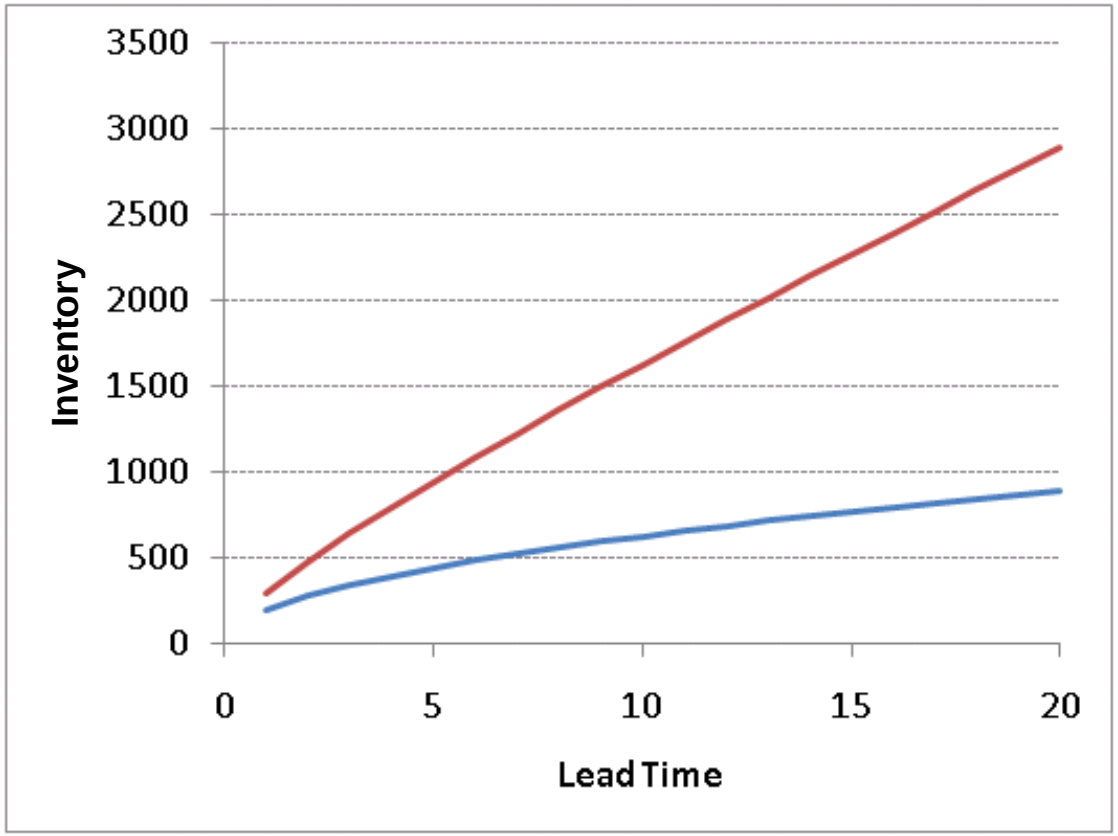
Lead time and safety stock



Reducing the lead time reduces safety inventory, especially if the target service level is high.

The impact of lead time on expected safety stock for four in-stock targets, 99.95%, 99.5%, 99.0% and 98%, top curve to bottom curve respectively. Demand in one period is Normally distributed with mean 100 and standard deviation 60

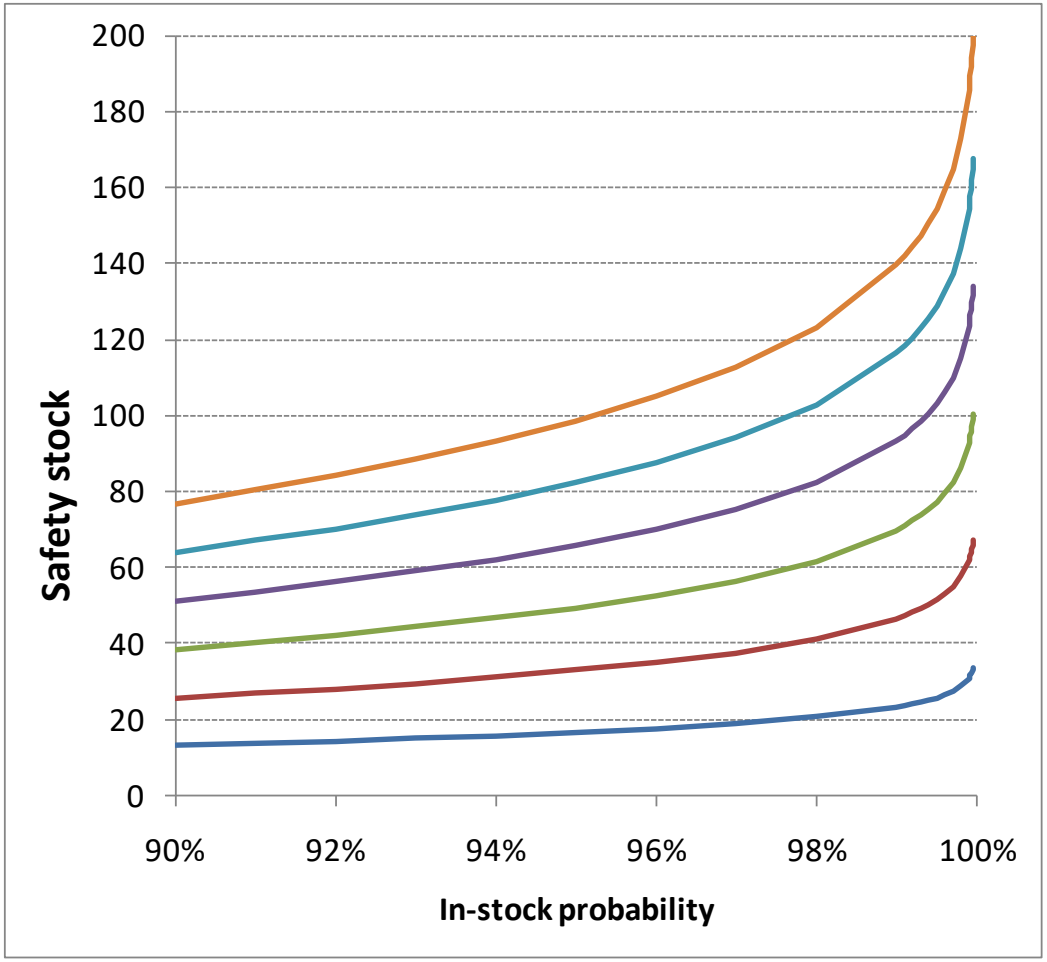
Lead time and pipeline inventory



Reducing the lead time might have an even more dramatic impact on pipeline inventory.

Expected safety stock (blue) and total inventory (red), which is expected on-hand inventory plus pipeline inventory, for a 99.95% type-I service level and demand in one period being Normally distributed with mean 100 and standard deviation 60.

Service level and safety stock



Safety stock increases exponentially as the target service approaches 100%.

The tradeoff between inventory and type-I service level for Normally distributed demand with mean of 100 over (L+1) periods. The curves differ in the standard deviation of demand over (L+1) periods: 60, 50, 40, 30, 20, and 10 from top to bottom.

Order up-to model summary

- The order-up-to model is appropriate for products with random demand but many replenishment opportunities.
- Expected inventory and service are controlled via the order-up-to level:
  - The higher the order up-to level the higher the in-stock probability, but the greater the expected on-hand inventory.
- The key factors that determine the amount of inventory are…
  - The length of the replenishment lead time
  - The length of the review period
  - The desired in-stock probability
  - Demand uncertainty
- When inventory obsolescence is not an issue, the optimal in-stock probability is generally quite high.

What if we relax some of OUT model's assumptions?

Demand / Customer

- Stationary demand
- Independent across periods
- Known parameters
- If stocked out, backorder

Supplier

- Lead time: deterministic and integer multiplier of review period
- Unlimited capacity
- Perfect quality

Costs & Environment

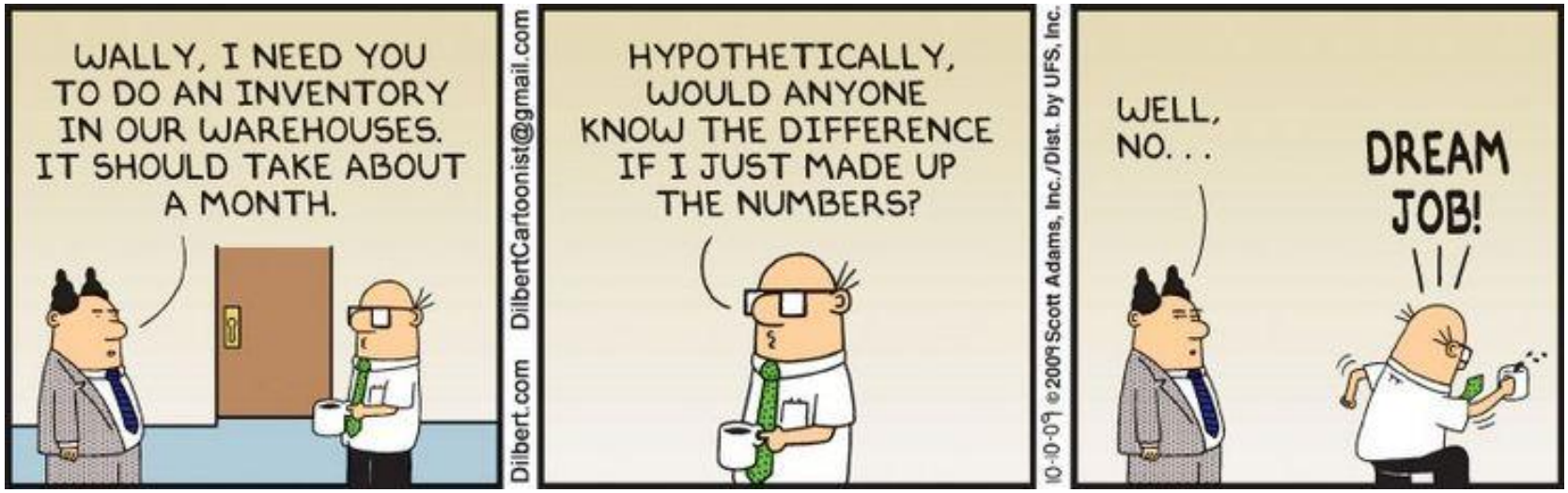
- Costs are known
- No set-up costs
- Long shelf-life
- No obsolescence

Inventory Policy

- Constant order-up-to point
- Constant review period
- Many replenishments over a long planning horizon

Readings

- Chapter 14 from Cachon & Terwiesch, 3rd edition



Thank you!

BACK-UP

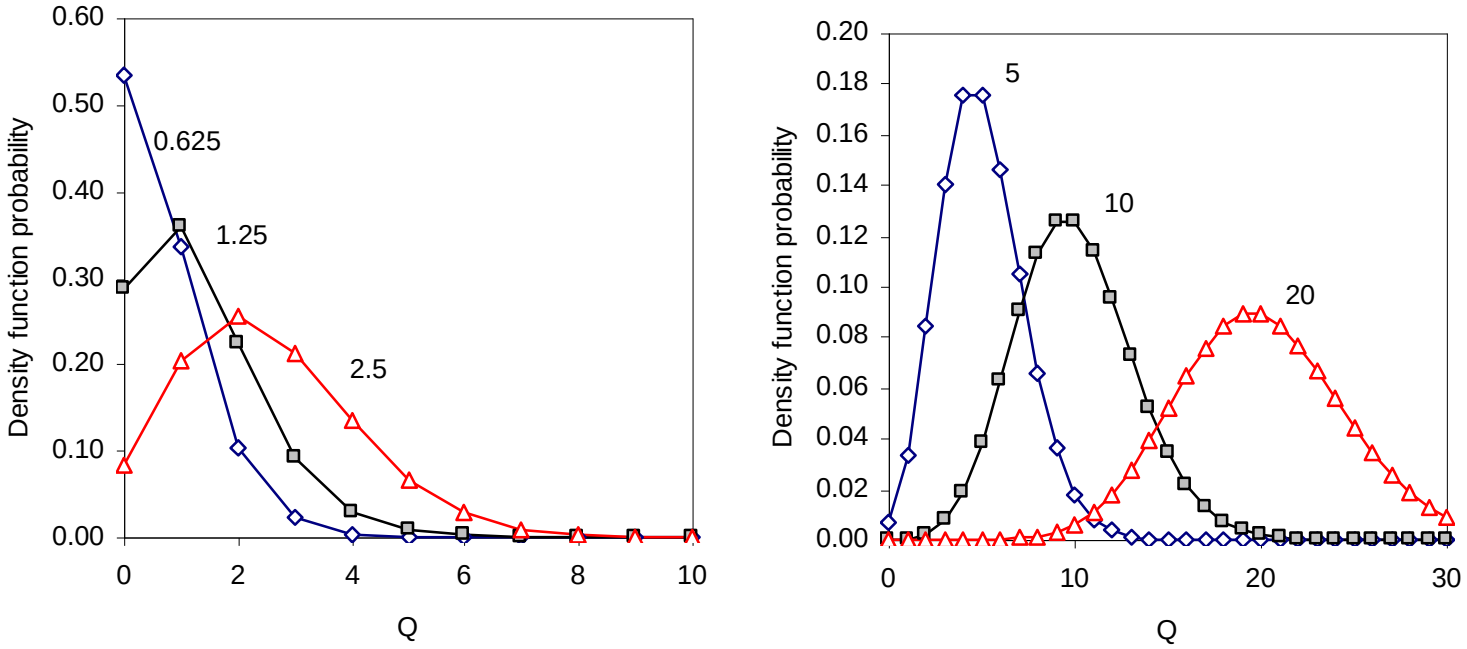
The Poisson Distribution

Poisson distribution

- The outcome of a random variable with a Poisson distribution is discrete (0,1,2,3,…) and positive (i.e., no negative outcomes).
- The Poisson is characterized by a single parameter, its mean:
  - Recall, the Normal distribution is defined by two parameters, the mean and standard deviation.
  - The standard deviation of a Poisson distribution is equal to the square root of its mean.
- The Poisson is related to the exponential distribution:
  - If the inter-arrival times of customers are exponentially distributed, then the number of customer arrivals in a fixed time interval is Poisson distributed.
- The Poisson is ideal for describing the demand of slow moving products, e.g., products that have average sales of 10 or fewer units over a particular period of time.

The shape of the Poisson distribution

- The Poisson's density function takes on different shapes, depending on its mean
  - With a large mean (more than 10) the Poisson takes on a "bell shape"
  - With a small mean (less than 5) the Poisson takes on different shapes that slowly change in the direction of a bell curve



Density function of 6 Poisson distributions with means 0.625, 1.25, 2.5, 5, 10 and 20

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Poisson's Distribution and Loss Functions

- There is no equivalent of "z-statistic" for the Poisson distribution.
  - You need to compute the distribution and loss function.

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$L(Q) = \lambda - \lambda \cdot F(Q-1) - Q \cdot (1 - F(Q))$$

- Consider the density function, f(Q), distribution function, F(Q), and the loss function, L(Q) of the Poisson with **mean 1.25** in the table to the right:
  - What is the probability that demand is exactly 2? 22.38%
  - What is the probability that demand is no more than 3? 96.17%
  - If you have 2 units to sell, then what is expected lost sales? 0.1811

Q	$f(Q)$	$F(Q)$	$L(Q)$
0	0.286505	0.286505	1.25000
1	0.358131	0.644636	0.53650
2	0.223832	0.868468	0.18114
3	0.093263	0.961731	0.04961
4	0.029145	0.990876	0.01134
5	0.007286	0.998162	0.00221
6	0.001518	0.999680	0.00038
7	0.000271	0.999951	0.00006
8	0.000042	0.999993	0.00001
9	0.000006	0.999999	0.00000
10	0.000001	1.000000	0.00000